Boundary-Aware Calibration of MAAT Structural Weights

A Reproducible Benchmark for Constraint-Dominated Structural Selection

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Abstract

We present a boundary-aware calibration of structural weights in the MAAT framework. The benchmark asks whether explicit constraint-boundary states alter the inferred hierarchy of structural selection sectors. To test this, we combine SAT hardness instances, unconstrained MAAT-Core optimisation states, and safe, near-boundary, and violated configurations into a fused defect ensemble of $N = 3400$ samples.

Fitting a closed MaxEnt structural functional yields a stable hierarchy. Robustness (*Respect*, R) emerges as the dominant contribution once explicit constraint violations and near-boundary regimes are included:

$$R > V > H > S \approx B$$

in absolute weight, with R carrying approximately 39.2% of the total normalised contribution.

The result supports a cautious but sharp conclusion: in this closed benchmark, structural selection is not driven primarily by dynamical complexity or entropy-like activity. It is dominated by constraint geometry, boundary stress, and the availability of stable margins.

1 Status and Scope

This paper is a numerical calibration note. It does not claim that the MAAT weights are universal constants. Instead, it tests whether the structural weights λ_a can be estimated in a closed reproducible system once explicit boundary information is included.

The central question is:

> *Does the inclusion of safe, near-boundary, and violated regimes change the inferred hierarchy of structural selection weights?*

2 Structural Functional

We consider a structural selection functional of the form

$$\mathcal{F}_{\text{struct}}[X] = \sum_{a \in \{H,B,S,V,R\}} \lambda_a d_a[X], \tag{1}$$

where $d_a[X] \geq 0$ are sector defects and $\lambda_a > 0$ are structural weights. The induced selection distribution is

$$P_\lambda[X_i] = \frac{\exp[-\mathcal{F}_{\text{struct}}[X_i]]}{\sum_j \exp[-\mathcal{F}_{\text{struct}}[X_j]]}. \tag{2}$$

The sectors are:

- H: equation or internal consistency,
- B: constraint balance,
- S: dynamical or generative activity,
- V: correlation, coupling, or connectedness,
- R: robustness, safety margin, or constraint respect.

Previous papers in this series used the nonlinear support form

$$\mathcal{F}^{\log}_{\text{struct}}[X] = -\sum_a \lambda_a \log(\epsilon + \Gamma_a[X]), \qquad \Gamma_a[X] = \frac{1}{1 + d_a[X]}. \tag{3}$$

For the purpose of weight calibration, we employ the linear proxy in Eq. (1). This proxy is monotonically related to the sector defects entering Eq. (3) and corresponds to the small-defect or chosen-cost representation of the same structural ordering. The simplification makes the MaxEnt fit transparent and reproducible while preserving the essential ranking information used in this benchmark.

3 Dataset Construction

The fused dataset is constructed from three sources:

Table 1: Fused defect dataset used for boundary-aware calibration.

Source	Samples	Description
SAT hardness atlas	2000	combinatorial constraint instances
MAAT-Core	500	unconstrained optimisation states
MAAT-Core boundary	900	safe, near-boundary, and violated regimes
Total	3400	heterogeneous closed calibration ensemble

The boundary dataset explicitly contains three classes:

- safe states with positive margins,
- near-boundary states with critical margins,
- violated states with negative margins.

The fused CSV file contains the common columns

$$(d_H, d_B, d_S, d_V, d_R, \text{label}, \text{source}).$$

4 Closed Lambda Fit

This calibration extends the previous MaxEnt weight estimate used in the effective-constants benchmark by adding explicit boundary regimes. The previous estimate enters only as a weak prior; the present fit is driven by the fused SAT, MAAT-Core, and boundary ensemble.

The weights are estimated by minimising a regularised MaxEnt matching loss. For an ensemble with defect matrix $d_a(X_i)$, define the expectation

$$\mathbb{E}_\lambda[d_a] = \sum_i P_\lambda[X_i]\, d_a(X_i). \tag{4}$$

The fitted weights minimise

$$\mathcal{L}(\lambda) = \sum_a (\mathbb{E}_\lambda[d_a] - d_a^*)^2 + \eta_{\text{bal}} \left\| \frac{\lambda}{\sum_b \lambda_b} - \frac{1}{5}\mathbf{1} \right\|^2 + \eta_{\text{prior}} \left\| \log \lambda - \log \lambda^{(0)} \right\|^2. \tag{5}$$

Here d_a^* are target defects determined from the best 20% of the fused ensemble by mean defect load. The weak prior $\lambda^{(0)}$ is the previous MaxEnt calibration used in the constants benchmark, and the balance term prevents collapse onto a single sector. Both regularisers are needed for different reasons: the balance term prevents the fit from becoming artificially one-dimensional, while the prior term keeps the calibration connected to the earlier MaxEnt estimate without fixing the new hierarchy in advance. The prior values are

$$\lambda^{(0)} = (1.610873,\ 2.428297,\ 4.623655,\ 4.644912,\ 8.832344)$$

for (H, B, S, V, R), respectively.

5 Results

The fit converges successfully with

$$\mathcal{L} = 0.0029543310, \qquad \sum_a \lambda_a = 20.62085487.$$

Table 2: Closed boundary-aware MAAT weight calibration.

Sector	λ_a	Share	Target d_a^*	Expected $\mathbb{E}_\lambda[d_a]$
H	3.218288	0.1561	0.192248	0.197083
B	2.469024	0.1197	0.199984	0.192900
S	2.526500	0.1225	0.310039	0.295512
V	4.328860	0.2099	0.243131	0.232882
R	8.078183	0.3917	0.010220	0.036344

The inferred hierarchy is

$$R > V > H > S \approx B.$$

The most important qualitative result is that R carries almost 40% of the total normalised weight.

6 Constraint Dominance

The dominant contribution is

$$\lambda_R \gg \lambda_H, \lambda_B, \lambda_S,$$

with

$$\frac{\lambda_R}{\lambda_B} \approx 3.27, \qquad \frac{\lambda_R}{\lambda_S} \approx 3.20.$$

This indicates that structural selection in the fused system is primarily controlled by robustness against constraint violation and boundary stress.

The R target is not matched exactly:

$$d_R^* = 0.010220, \qquad \mathbb{E}_\lambda[d_R] = 0.036344.$$

This residual mismatch is informative. It shows that the fused ensemble contains genuine boundary tension: the optimiser strongly increases λ_R, but the available configurations cannot fully suppress the robustness defect without compromising the remaining sectors. Thus the mismatch is not merely numerical error; it is evidence that the boundary sector introduces a real tension between robustness and the other structural requirements in the fused ensemble.

7 Role of the Boundary Dataset

The role of the boundary dataset is essential:

- the SAT dataset contributes combinatorial constraint structure,
- the unconstrained MAAT-Core dataset contributes optimisation states,
- the boundary dataset supplies explicit margins, violations, and near-boundary stress.

Without explicit boundary information, the R sector is weakly identified. With boundary information, R becomes the dominant selector. This supports the interpretation that constraint geometry is not merely an auxiliary feature, but a central component of structural selection.

8 Figures

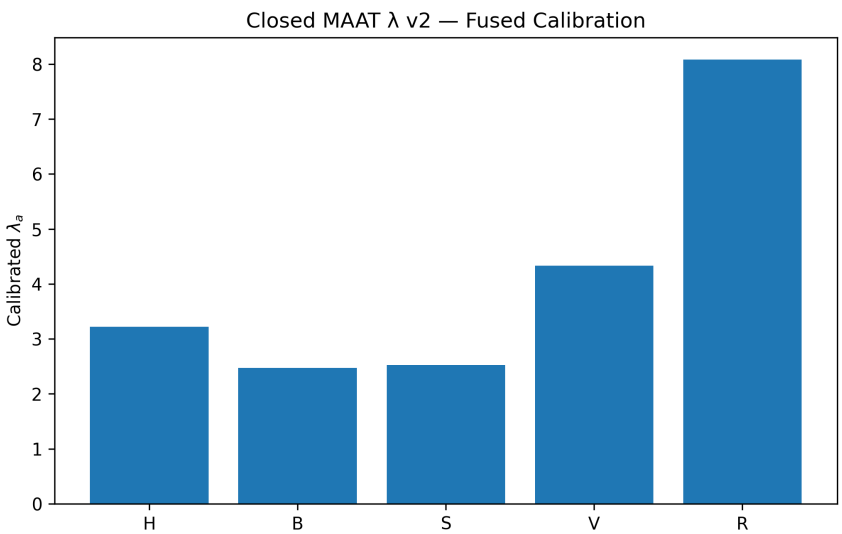

Figure 1: Calibrated structural weights λ_a for the fused SAT, MAAT-Core, and boundary dataset.

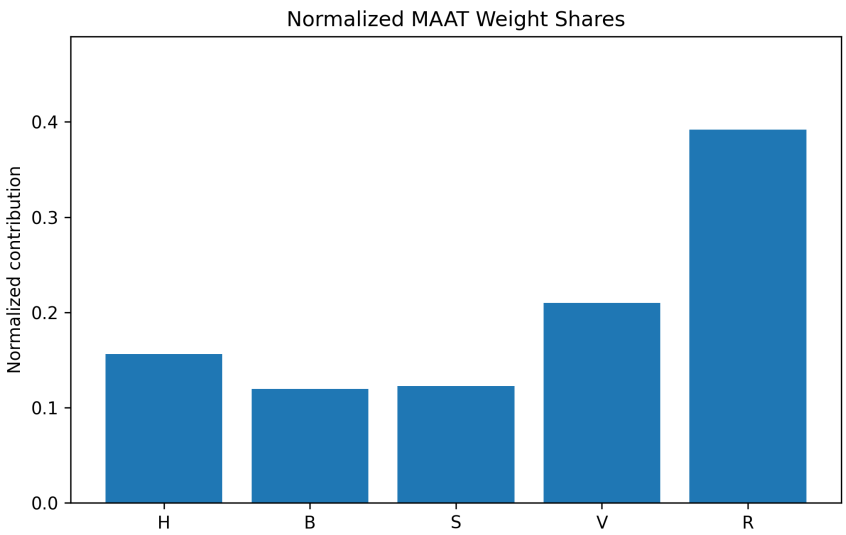

Figure 2: Normalised sector shares. The robustness sector R carries approximately 39.2% of the total calibrated weight.

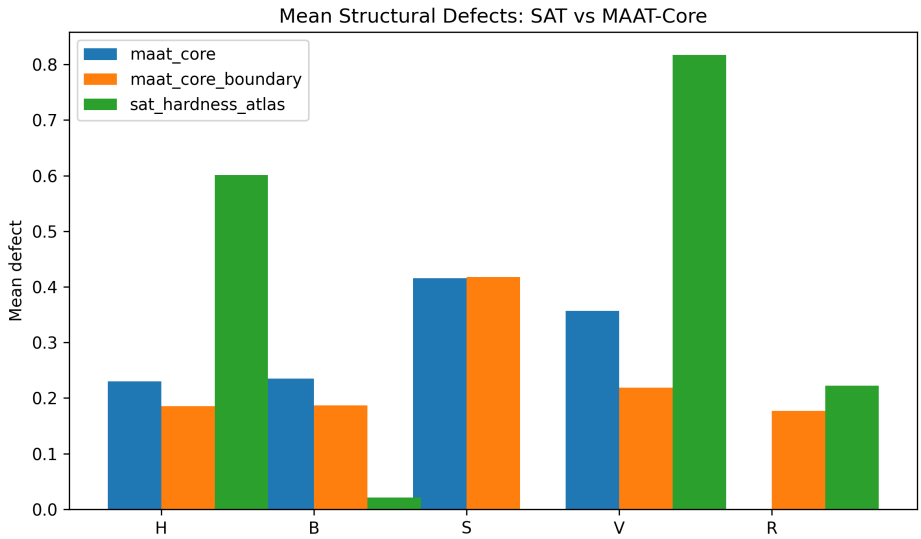

Figure 3: Mean structural defects by source. The fused ensemble combines SAT constraint structure, unconstrained MAAT-Core samples, and boundary-aware stress regimes.

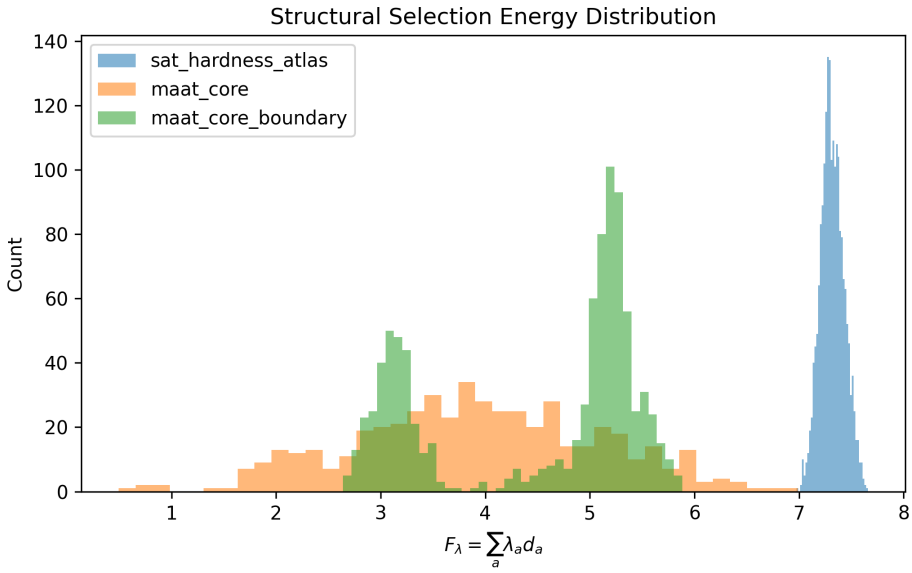

Figure 4: Distribution of calibrated structural energy $F_\lambda = \sum_a \lambda_a d_a$ across sources.

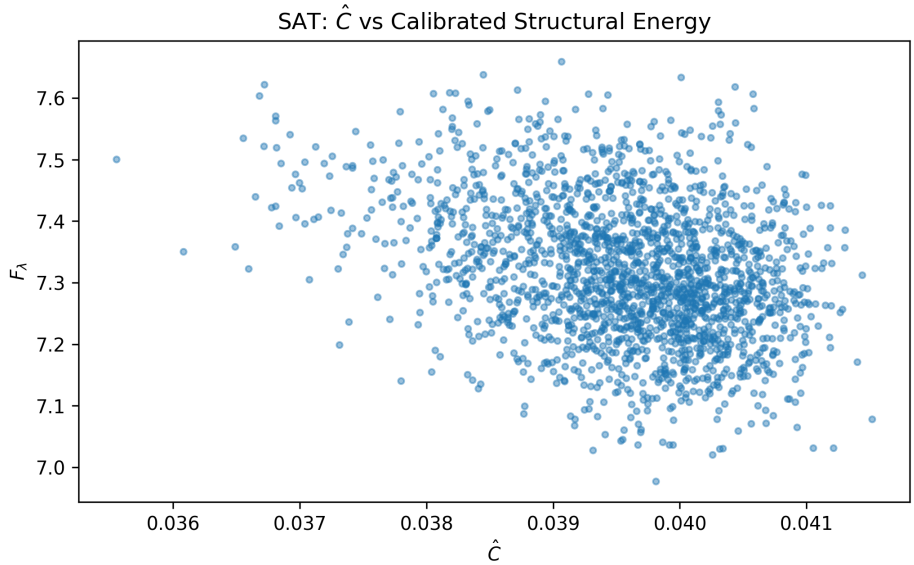

Figure 5: SAT subset: structural energy F_λ compared with the available structural hardness proxy $\hat{C}$.

9 Interpretation

The calibrated hierarchy suggests a structural ordering:

$$\text{robustness} \rightarrow \text{connectedness} \rightarrow \text{consistency, activity, and balance.}$$

Physically and computationally, this means:

- constraint geometry dominates phase selection,
- stability margins shape accessible configurations,

- near-boundary regimes act as critical regions,
- robustness is not an ethical decoration but an operational selector.

10 Limitations

The present benchmark remains limited in several ways:

- the boundary dataset is generated by MAAT-Core rather than measured from an external physical system,
- the fit includes a weak prior toward the previous MaxEnt calibration,
- the target defects are defined by a best-20% rule and may change under alternative selection criteria,
- the calibration is domain-fused and should not yet be interpreted as a universal set of MAAT constants.

11 Conclusion

We have shown that structural weights can be calibrated in a closed boundary-aware system. The inclusion of explicit constraint-boundary regimes changes the result: robustness becomes the dominant selection sector.

This supports the hypothesis that structural selection is governed not only by energy, entropy, or complexity, but by the geometry of constraints and the availability of stable margins.

Reproducibility

All code, data files, and generated plots for this benchmark are archived in the public repository:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/boundary_aware_lambda_calibration

The numerical pipeline consists of:

- `export_sat_defects.py`,
- `export_core_defects.py`,
- `export_core_boundary_defects.py`,
- `merge_defects.py`,
- `fit_closed_maat_lambda_v1.py`,
- `plot_closed_maat_lambda_v2.py`.

The main analysis can be reproduced with:

```
python3 merge_defects.py
python3 fit_closed_maat_lambda_v1.py
python3 plot_closed_maat_lambda_v2.py
```

The main data files are:

- `maat_defects_sat.csv`,

- `maat_defects_core.csv`,
- `maat_defects_core_boundary.csv`,
- `maat_defects_fused.csv`,
- `closed_maat_lambda_fit_results.json`.

References

[1] E. T. Jaynes, "Information Theory and Statistical Mechanics," *Physical Review* **106**, 620–630 (1957).

[2] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, Wiley, 2nd edition (2006).

[3] C. Krieg, *Structural Selection Principle*, ongoing paper series and code repository (2026).